Power Analysis Worksheet

Fillable PDF. Click a blue box to type; tick the small squares. Save with **File › Save** and attach the saved PDF to your assignment. Works best in the free Adobe Acrobat Reader; macOS Preview and web browsers also work. If an answer does not fit its box, continue it on an extra page. Source: `Step_09_Power_Analysis_Sampling_and_Pilots/power_analysis_template.md` (rebuild with `tools/make_fillable_pdfs.py`).

Complete this worksheet for your primary (confirmatory) hypothesis, then attach it together with the output of your power-analysis script (R or Python) to your proposal or pre-registration. Example R scripts for common designs are in `scripts/` (see the Step 09 README).

> **Rule of thumb for this worksheet:** every number must have a source. If an effect size comes from a published study, give the citation and DOI and show how you converted it. If it is the smallest effect you care about, say why that size matters for your question.

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1. Study design

Field	Your entry
Study title	
Primary hypothesis (H1)	
Primary outcome and how it is measured	
Design	☐ Two groups, between subjects ☐ BACI (before/after × control/impact) ☐ ACI (after only, control/impact) ☐ Within subjects ☐ Other:
Statistical test for H1	
Unit of analysis (person, household, site…)	
Allocation ratio (e.g., 1:1)	
Software and script used	☐ R ☐ Python ☐ G*Power ☐ Other:
Script file name	

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2. Assumed effect size

Field	Your entry
Effect size metric	☐ Cohen's d ☐ Cohen's f ☐ Correlation r ☐ Odds ratio / two proportions ☐ Other:
Assumed value	
Source (author, year, journal)	
DOI	

Field	Your entry
Statistic reported in the source (e.g., means and SDs, t, F, r)	
How you converted it to your metric (formula)	
Smallest effect size of interest (SESOI) and why it matters	
Why you did not rely on a pilot-study estimate alone	

Tip: pilot studies are too small to estimate an effect size precisely. Use them to test procedures, completion time, and the reliability of your measures; base the effect size on published evidence or on your smallest effect size of interest.

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3. Error rates and inflation for attrition

Field	Your entry
Alpha (Type I error rate)	
One- or two-sided test	☐ Two-sided ☐ One-sided (justify below)
Target power (1 − β)	
Required N from the power analysis (per group)	
Required N from the power analysis (total)	
Expected attrition between waves (%) and source	
Expected non-response / exclusions (%) and source	
Inflated N to recruit = N ÷ ((1 − attrition) × (1 − non-response))	

Justification for any non-default choice (e.g., one-sided test, alpha other than .05, power other than .80):

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4. Sensitivity analysis

Report the sample size your design needs across a plausible range of effect sizes, or the smallest effect your feasible N can detect.

Assumed effect size	Required N per group	Required N total	Notes

Need more rows? Add a continuation page or attach a spreadsheet with the same columns.

Smallest effect detectable with the N you can realistically recruit (at your alpha and power):

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5. Script output

Paste the key lines printed by your script (or attach the full output as a separate page).

Paste the output here.

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6. Checklist before you attach this worksheet

- ☐ The effect size has a real, checkable source (citation and DOI), or a stated smallest effect size of interest.
- ☐ I showed how a reported statistic was converted to d, f, or r.
- ☐ Alpha, sidedness, and power are stated and justified.
- ☐ The recruited N is inflated for attrition and non-response.
- ☐ A sensitivity analysis shows how N changes across plausible effect sizes.
- ☐ The script and its output are attached, with the package versions printed by the script.

Cross-reference: `README.md` · `scripts/` · `../Step_04_Study_Architecture_and_Experimental_Design/design_decision_matrix.md` · `../Step_05_Preregistration_and_Open_Science/preregistration_template.md`